

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	07/07/2026
Team ID	SWUID2026196344
Project Name	Nutrition Assistant
Maximum Marks	4 Marks

Technical Architecture:

The deliverable includes the architecture of the **Nutrition Assistant** system, illustrating how users interact with the web application, how requests are processed through the backend, and how data is securely stored in the database. The architecture follows a **three-tier MERN stack design** consisting of the presentation layer (React.js), application layer (Node.js and Express.js), and database layer (MongoDB). Tables 1 and 2 describe the technologies, components, and key architectural characteristics used in the application.

Example: Nutrition Assistant – Personalized Nutrition Management System

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>

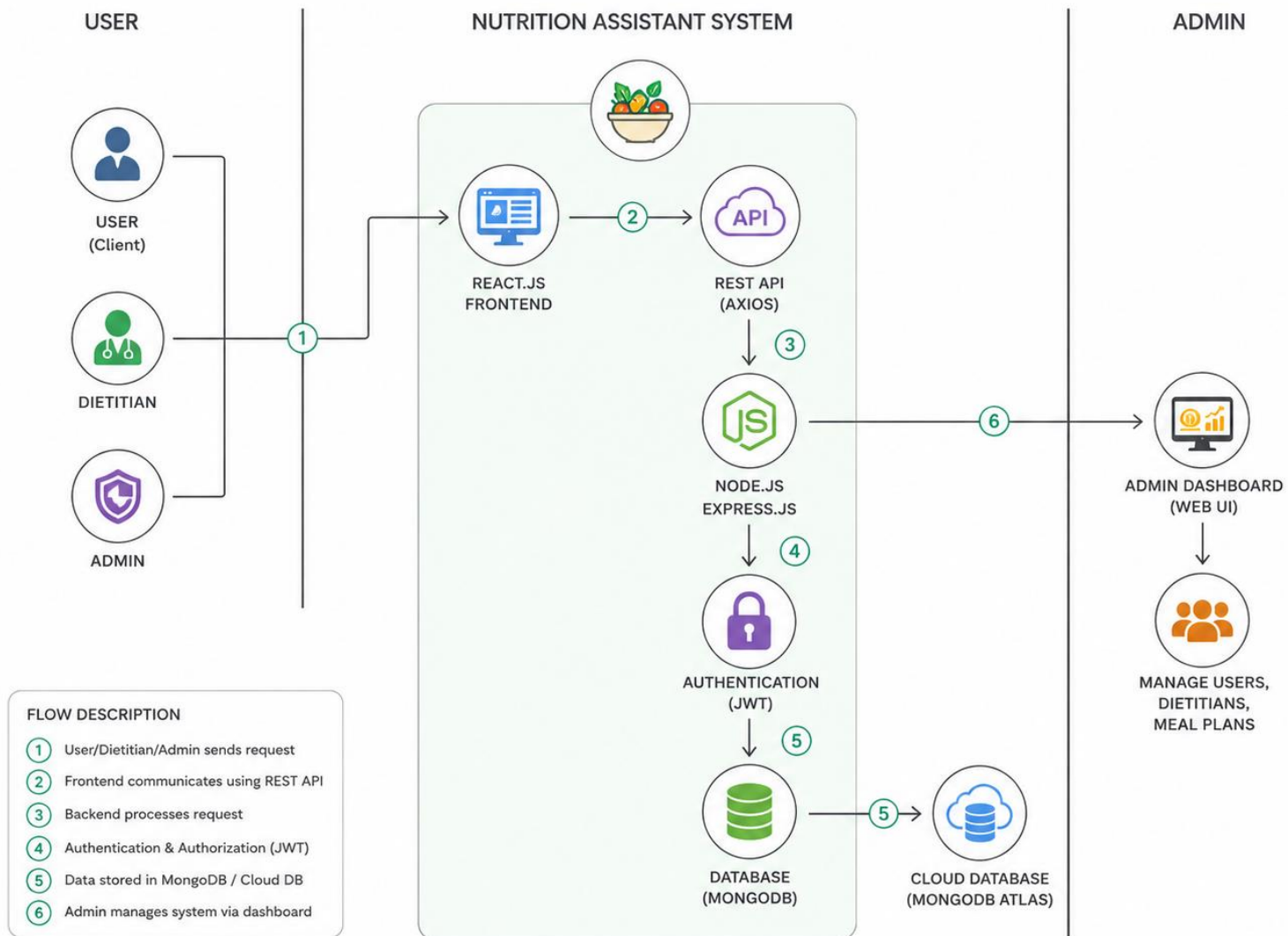


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Responsive web application for users, dietitians, and admin	HTML, CSS, JavaScript / Angular Js / React Js etc.
2.	Application Logic-1	User authentication and authorization	Node.js, Express.js, JWT, bcrypt
3.	Application Logic-2	Meal plan management and nutrition tracking	Node.js, Express.js
4.	Application Logic-3	Progress monitoring and admin management	Node.js, Express.js
5.	Database	Stores users, meal plans, nutrition records, and progress	MongoDB, Mongoose
6.	Cloud Database	Cloud-hosted database (optional)	MongoDB Atlas
7.	File Storage	Profile images and application files	Local File System
8.	External API-1	HTTP communication between frontend and backend	Axios
9.	External API-2	API testing and validation	Postman
10.	Machine Learning Model	Not Applicable	N/A
11.	Infrastructure (Server / Cloud)	Local development and deployment	Local Server (Node.js), MongoDB Atlas

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frontend and backend development frameworks	React.js, Node.js, Express.js, Bootstrap
2.	Security Implementations	Secure authentication, password encryption, and access control	JWT, bcrypt, Express Middleware

S.No	Characteristics	Description	Technology
3.	Scalable Architecture	Three-tier architecture with REST APIs	MERN Stack
4.	Availability	Supports continuous access through web browser	Node.js Server, MongoDB Atlas
5.	Performance	Fast API responses and optimized database operations	MongoDB Indexing, Axios, React Optimization

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>